Chapter 24 Check

Contents

24 Historical Introduction

The history of supersymmetry is as peculiar as anything in the history of science. Suggested in the early 1970s, supersymmetry has been elaborated since then into a beautiful mathematical formalism that unites particles of different spin into symmetry multiplets and has profound implications for fundamental physics. Yet there is so far not a shred of direct experimental evidence and only a few bits of indirect evidence that supersymmetry has anything to do with the real world. If (as I expect) supersymmetry does turn out to be relevant to nature, it will represent a striking success of purely theoretical insight.

Chapter 25 will begin the construction of supersymmetry theories from first principles. In the present chapter we shall introduce supersymmetry along chronological rather than logical lines.

24.1 Unconventional Symmetries and 'No-Go' Theorems

In the early 1960s the symmetry $SU(3)$ of Gell-Mann and Ne'eman (discussed in Section 19.7) successfully explained the relations between various strongly interacting particles of different charge and strangeness but of the same spin. The idea then grew up that perhaps $SU(3)$ is part of a larger symmetry, which has the unconventional effect of uniting $SU(3)$ multiplets of different spin [1]. There is such an approximate symmetry in the non-relativistic quark model, under $SU(6)$ transformations on quark spins and flavors, analogous to an earlier $SU(4)$ symmetry of nuclear physics that had been introduced in 1937 by Wigner [2]. As described in detail in Appendix A of this chapter, this $SU(6)$ symmetry unites the pseudoscalar meson octet π, K, $\bar{K}$, and η, the vector meson octet ρ, K^*, $\bar{K}^*$, and ω, and the vector meson singlet ϕ in a single 35 multiplet, and also unites the spin 1/2 baryon octet N, Σ, Λ, and Ξ with the spin 3/2 baryon decuplet Δ, $\Sigma(1385)$, $\Xi(1530)$, and Ω in a single 56 multiplet. The $SU(6)$ symmetry scored a number of successes, but it is actually nothing but a consequence of the approximate spin and flavor independence of forces in the quark model; the $SU(6)$ symmetry is somewhat weaker than the assumption of spin and flavor independence, but, as shown in Appendix A, there is no evidence that the predictions of $SU(6)$ symmetry are any more accurate than those of complete spin and flavor independence.

Nevertheless, there were various attempts to generalize the $SU(6)$ symmetry of the non-relativistic quark model to a fully relativistic quantum theory [3]. These attempts all failed, and a number of authors showed under various restrictive assumptions that this is in fact impossible [4]. The most far-reaching theorem of this sort was proved in 1967 by Coleman and Mandula [5]. They adopted reasonable assumptions about the finiteness of the number of particle types below any given mass, the existence of scattering at almost all energies, and the analyticity of the S-matrix, and used them to show that the most general Lie algebra of symmetry operators that commute with the S-matrix, that take single-particle states into single-particle states, and that act on multiparticle states as the direct sum of their action on single-particle states consists of the generators P_μ and $J_{\mu\nu}$ of the Poincaré group, plus ordinary internal symmetry generators that act on one-particle states with matrices that are diagonal in and independent of both momentum and spin. We will use this theorem as an essential ingredient in our analysis of all possible supersymmetry algebras in four spacetime dimensions in Chapter 25, and in higher spacetime dimensions in Chapter 32. In Section 32.3 we will consider supersymmetry algebras in theories that involve extended objects, for which the Coleman–Mandula theorem does not apply.

Coleman and Mandula's proof is ingenious and complicated. A version is presented in Appendix B to this chapter. In the present section we shall give a very simple purely kinematic proof of one piece of this theorem, but this piece is enough to show clearly why an unconventional symmetry like $SU(6)$ is possible in non-relativistic but not in relativistic theories. We shall use Lorentz invariance to show that if the Lie algebra of all symmetry operators B_α that commute with the momentum generators P_μ consists of the P_μ themselves plus the Hermitian generators B_A of some finite-parameter semi-simple compact[1] Lie subalgebra $\mathcal{A}$, then the B_A must be the generators of an ordinary internal symmetry, in the sense that they act on single-particle states with matrices that are diagonal in and independent of both momentum and spin. No use is made in this theorem of the properties of the S-matrix, of the finiteness of the particle spectrum, or of assumptions about how the symmetry generators act on physical states. The Lie algebra of $SU(6)$ is of course both semi-simple and compact, so this theorem rules out the use of any such symmetry in relativistic theories to derive relations among particles of different spin.

Here is the proof. Let all symmetry generators that commute with the four-momentum P_μ form a Lie algebra spanned by the generators B_α. Consider the effect on these generators of a proper Lorentz transformation $x^\mu \longrightarrow \Lambda^\mu{}_\nu x^\nu$, which is represented on Hilbert space by the unitary operator $U(\Lambda)$. It is easy to see that the operator $U(\Lambda)B_\alpha U^{-1}(\Lambda)$ is a Hermitian symmetry generator that commutes with $\Lambda_\mu{}^\nu P_\nu$, so since $\Lambda_\mu{}^\nu$ is non-singular, this operator must commute with P_μ, and therefore must be a linear combination of the B_α:

$$U(\Lambda)B_\alpha U^{-1}(\Lambda) = \sum_\beta D^\beta{}_\alpha(\Lambda)B_\beta\,, \tag{24.1.1}$$

[1]For the definition of semi-simple and compact Lie algebras, see the footnote in Section 15.2.

with $D^\beta{}_\alpha(\Lambda)$ a set of real coefficients that furnish a representation of the homogeneous Lorentz group

$$D(\Lambda_1)D(\Lambda_2) = D(\Lambda_1\Lambda_2)\,. \tag{24.1.2}$$

Further, the $U(\Lambda)B_\alpha U^{-1}(\Lambda)$ satisfy the same commutation relations as the B_α, so the structure constants $C^\gamma{}_{\alpha\beta}$ of this Lie algebra are invariant tensors in the sense that

$$C^\gamma{}_{\alpha\beta} = \sum_{\alpha'\beta'\gamma'} D^{\alpha'}{}_\alpha(\Lambda)D^{\beta'}{}_\beta(\Lambda)D^\gamma{}_{\gamma'}(\Lambda^{-1})C^{\gamma'}{}_{\alpha'\beta'}\,. \tag{24.1.3}$$

Contracting this with the corresponding equation for $C^\alpha{}_{\gamma\delta}$, we find that

$$g_{\beta\delta} = \sum_{\beta'\delta'} D^{\beta'}{}_\beta(\Lambda)D^{\delta'}{}_\delta(\Lambda)g_{\beta'\delta'}\,, \tag{24.1.4}$$

where $g_{\beta\delta}$ is the Lie algebra metric

$$g_{\beta\delta} \equiv \sum_{\alpha\gamma} C^\gamma{}_{\alpha\beta}C^\alpha{}_{\gamma\delta}\,. \tag{24.1.5}$$

Because all of these generators commute with P_μ, we have $C^\alpha{}_{\mu\beta} = -C^\alpha{}_{\beta\mu} = 0$, so $g_{\mu\alpha} = g_{\alpha\mu} = 0$.

We will distinguish the symmetry generators other than the P_μ by using subscripts A, B, etc. in place of α, β, etc. Using the vanishing of $C^A{}_{\mu B} = -C^A{}_{B\mu}$ in Eq. (24.1.5) gives $g_{AB} = \sum_{CD} C^D{}_{AC}C^C{}_{BD}$. We assumed that the generators B_A span a compact semi-simple Lie algebra, so the matrix g_{AB} is positive-definite. Eqs. (24.1.4) and (24.1.2) show that the matrices $g^{1/2}D(\Lambda)g^{-1/2}$ furnish a real orthogonal and hence unitary finite-dimensional representation of the homogeneous Lorentz group. But because the Lorentz group is non-compact, *the only such representation is the trivial one*, for which $D(\Lambda) = \mathbf{1}$. (This is the place where relativity makes all the difference: the semi-simple part of the Galilean group is the *compact* group $SU(2)$, which of course has an infinite number of unitary finite-dimensional representations.) With $D(\Lambda) = \mathbf{1}$, the generators B_A commute with $U(\Lambda)$ for all Lorentz transformations $\Lambda^\mu{}_\nu$.

Acting on the state $\|p, n\rangle$ of a single stable particle with momentum p^μ and spin and species labelled by a discrete index n, an operator like B_A that commutes with P_μ can only yield a linear combination of such states

$$B_A\|p, n\rangle = \sum_m \big(b_A(p)\big)_{mn}\|p, m\rangle\,. \tag{24.1.6}$$

The fact that the B_A commute with what we called 'boosts' in Section 2.5 implies that the $b_A(p)$ are independent of momentum, and the fact that the B_A commute with rotations implies that the $b_A(p)$ act as unit matrices on spin indices, so the B_A are the generators of an ordinary internal symmetry, as was to be proved.

24.2 The Birth of Supersymmetry

If theoretical physics followed logic in its evolution, then after the proof of the Coleman–Mandula theorem someone seeking exceptions to this theorem would have noticed that it deals only with transformations that take bosons into bosons and fermions into fermions and are therefore generated by operators that satisfy commutation relations rather than anticommutation relations. This would have raised the question of whether a relativistic theory can have symmetries acting non-trivially on particle spins that take fermions and bosons into each other, and that therefore satisfy anticommutation relations rather than commutation relations. Exploring the possible structure of such a superalgebra along the lines described in the following chapter, supersymmetry would have emerged as the only possibility.

This is not what happened. Instead, supersymmetry arose in a series of articles on string theory and independently in a pair of little-noticed papers, about which more later, none of which show any sign that the authors were at all concerned with the Coleman–Mandula theorem.

Starting in the late 1960s, the effort to construct S-matrix elements for strong-interaction processes that would satisfy various theoretical requirements led to a new picture of various types of hadrons, as different modes of vibration of a string [6]. A point on a string labelled by a parameter σ will at time τ on some fixed clock have spacetime coordinates $X^\mu(\sigma,\tau)$, so the theory of a string's motion in d spacetime dimensions may be regarded as a two-dimensional field theory with d bosonic fields, with action

$$\begin{aligned} I[X] &= \frac{T}{2}\int d\sigma \int d\tau\, \eta_{\mu\nu}\left[\frac{\partial X^\mu}{\partial\tau}\frac{\partial X^\nu}{\partial\tau} - \frac{\partial X^\mu}{\partial\sigma}\frac{\partial X^\nu}{\partial\sigma}\right] \\ &= T\int d\sigma^+ \int d\sigma^-\, \eta_{\mu\nu}\frac{\partial X^\mu}{\partial\sigma^+}\frac{\partial X^\nu}{\partial\sigma^-}, \end{aligned} \tag{24.2.1}$$

where T is a constant known as the string tension; $\mu = 0, 1, \ldots, d-1$; and $\sigma^\pm$ are the two-dimensional 'light-cone' coordinates $\sigma^\pm \equiv \tau \pm \sigma$. This action can be derived from a more general version, with complete invariance[2] under transformations of a pair of 'worldsheet coordinates' σ_k

$$I[X] = -\frac{T}{2}\int d^2\sigma\, \eta_{\mu\nu}\sqrt{\mathrm{Det}\, g}\, g^{kl}\frac{\partial X^\mu}{\partial\sigma^k}\frac{\partial X^\nu}{\partial\sigma^l}, \tag{24.2.2}$$

by passing to a special coordinate system in which the worldsheet metric g_{kl} satisfies the condition

$$\sqrt{\mathrm{Det}\, g}\, g^{kl} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{24.2.3}$$

In much the same way that in electrodynamics the problems introduced by the negative sign of the action for timelike photons are eliminated by the gauge invariance of the theory, here the problems introduced by the negative sign of $\eta_{\mu\nu}$ in Eqs. (24.2.1) and (24.2.2) for $\mu = \nu = 0$ are eliminated by the invariance of the action (24.2.2) (for appropriate boundary conditions) under general transformations of the worldsheet coordinates. In the special coordinate system in which the action takes the form (24.2.1), there is an important remnant of invariance under general worldsheet coordinate transformations: invariance under the

[2] This symmetry is violated by quantum anomalies, like those discussed in Chapter 22, except in $d = 26$ spacetime dimensions for the purely bosonic theory, or $d = 10$ dimensions after the introduction of fermions.

global *conformal transformations*:

$$\sigma^{\pm} \longrightarrow f^{\pm}(\sigma^{\pm}), \tag{24.2.4}$$

with $f^{\pm}$ a pair of independent arbitrary functions.

The particles described by this string theory do not match those seen in the real world. Ramond [7] and Neveu and Schwarz [8] in 1971, aiming respectively to introduce particles with half-integer spin or with the quantum numbers of pions, suggested the addition of d fermionic field doublets $\psi_1^\mu(\sigma,\tau)$ and $\psi_2^\mu(\sigma,\tau)$. Shortly after, Gervais and Sakita [9] introduced an action for this theory:

$$I[X,\psi] = \int d\sigma^+ \int d\sigma^- \left[T \frac{\partial X^\mu}{\partial \sigma^+}\frac{\partial X_\mu}{\partial \sigma^-} + i\psi_2^\mu \frac{\partial}{\partial \sigma^+}\psi_{2\mu} + i\psi_1^\mu \frac{\partial}{\partial \sigma^-}\psi_{1\mu}\right], \tag{24.2.5}$$

and noted that conformal invariance could be maintained by extending the conformal transformations (24.2.4) to act also on the fermion fields

$$\psi_1^\mu \longrightarrow \left(\frac{df^+}{d\sigma^+}\right)^{-1/2} \psi_1^\mu, \qquad \psi_2^\mu \longrightarrow \left(\frac{df^-}{d\sigma^-}\right)^{-1/2} \psi_2^\mu. \tag{24.2.6}$$

Gervais and Sakita pointed out that, in addition to two-dimensional conformal invariance and d-dimensional Lorentz invariance, for appropriate boundary conditions this theory has a symmetry under infinitesimal transformations that interchange the bosonic field X^μ with the fermionic fields ψ_r^μ:

$$\begin{aligned} \delta\psi_2^\mu(\sigma^+,\sigma^-) &= iT\alpha_2(\sigma^-)\frac{\partial}{\partial\sigma^-}X^\mu(\sigma^+,\sigma^-), \\ \delta\psi_1^\mu(\sigma^+,\sigma^-) &= iT\alpha_1(\sigma^+)\frac{\partial}{\partial\sigma^+}X^\mu(\sigma^+,\sigma^-), \\ \delta X^\mu(\sigma^+,\sigma^-) &= \alpha_2(\sigma^-)\psi_2^\mu(\sigma^+,\sigma^-) + \alpha_1(\sigma^+)\psi_1^\mu(\sigma^+,\sigma^-), \end{aligned} \tag{24.2.7}$$

where α_1 and α_2 are a pair of infinitesimal *fermionic* functions of σ^+ and σ^-, respectively, like the Grassmann variables introduced in Section 9.5. This was an example of what has subsequently come to be called supersymmetry, a symmetry connecting bosons and fermions, but thus far it was only a symmetry of a two-dimensional field theory, not of a physical theory in four spacetime dimensions.

A few years later Wess and Zumino [10] referred back to the example of supersymmetry that had been provided by References [7], [8], and [9], and commented that it would be natural to try to extend the idea of supersymmetry to quantum field theories in *four* spacetime dimensions. They constructed several supersymmetric models. The simplest involved a single Majorana (self-charge-conjugate Dirac) field ψ, a pair of real scalar and pseudoscalar bosonic fields A and B, and a pair of real scalar and pseudoscalar bosonic auxiliary fields F and G, with invariance under the infinitesimal transformation[3]

$$\begin{aligned}
&\delta A = (\bar{\alpha}\psi), && \delta B = -i(\bar{\alpha}\gamma_5\psi), \\
&\delta\psi = \partial_\mu(A - i\gamma_5 B)\gamma^\mu\alpha + (F - i\gamma_5 G)\alpha, && \\
&\delta F = (\bar{\alpha}\gamma^\mu\partial_\mu\psi), && \delta G = -i(\bar{\alpha}\gamma_5\gamma^\mu\partial_\mu\psi),
\end{aligned} \tag{24.2.8}$$

where α is an arbitrary constant infinitesimal Majorana fermion c-number parameter. If we require invariance of the action under these transformations, then the most general real, Lorentz-invariant, parity-conserving, renormalizable Lagrangian density built out of these ingredients is

$$\begin{aligned}
\mathcal{L} = &-\frac{1}{2}\partial_\mu A\,\partial^\mu A - \frac{1}{2}\partial_\mu B\,\partial^\mu B - \frac{1}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi \\
&+\frac{1}{2}(F^2 + G^2) + m\left[FA + GB - \frac{1}{2}\bar{\psi}\psi\right] \\
&+ g\left[F(A^2 + B^2) + 2GAB - \bar{\psi}(A + i\gamma_5 B)\psi\right].
\end{aligned} \tag{24.2.9}$$

Since the auxiliary fields F and G enter quadratically, we can derive an equivalent Lagrangian by setting them equal to the values given by the field equations

$$F = -mA - g(A^2 + B^2), \qquad G = -mB - 2gAB. \tag{24.2.10}$$

The Lagrangian density then becomes

$$\begin{aligned}
\mathcal{L} = &-\frac{1}{2}\partial_\mu A\,\partial^\mu A - \frac{1}{2}\partial_\mu B\,\partial^\mu B - \frac{1}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi \\
&-\frac{1}{2}m^2\left[A^2 + B^2\right] - \frac{1}{2}m\bar{\psi}\psi \\
&- gmA(A^2 + B^2) - \frac{1}{2}g^2(A^2 + B^2)^2 - g\bar{\psi}(A + i\gamma_5 B)\psi.
\end{aligned} \tag{24.2.11}$$

This Lagrangian density exhibits relations not only between scalar and fermion masses, but also between Yukawa interactions and scalar self-couplings, which are characteristic of supersymmetric theories. Wess and Zumino also described supersymmetry transformations and gave a Lagrangian for a supermultiplet containing a vector field. (We shall go into all this in more detail in Chapter 26.) Finally, in a second paper, Wess and Zumino [11] recalled the Coleman–Mandula theorem and traced the apparent violation of this theorem to the fact that the symmetry generators here satisfy anticommutation rather than commutation relations. It was a few more years before Gliozzi, Scherk and Olive [11a] showed that it was possible to construct a superstring theory with spacetime as well as worldsheet supersymmetry by imposing suitable periodicity conditions on the fields of the Ramond–Neveu–Schwarz model.

Unknown to Wess and Zumino, at the time of their first papers on supersymmetry in four spacetime dimensions this symmetry had already appeared in a pair of papers

[3]The notation for Dirac matrices used here is explained in the Preface and in Section 5.4. The γ_5 used here (which satisfies $\gamma_5^2 = \mathbf{1}$) is a factor of i times that used by Wess and Zumino, and the covariant conjugate $\bar{\psi}$ of any spinor ψ is defined here as i times that of Wess and Zumino. For this reason, some of the phases in Eqs. (24.2.8)–(24.2.10) are different from those in Reference [10].

published in the Soviet Union. In 1971 Gol'fand and Likhtman [12] had extended the algebra of the Poincaré group discussed in Section 2.4 to a superalgebra and used the requirement of invariance under this superalgebra to construct supersymmetric field theories in four spacetime dimensions. Their paper though prophetic gave few details, and was generally ignored until much later. Independently Volkov and Akulov [13] in 1973 discovered what today would be called spontaneously broken supersymmetry, but they used their formalism to identify the Goldstone fermion associated with supersymmetry breaking with the neutrino, an idea that met with no success. For most theorists, especially outside the Soviet Union, supersymmetry as a possible symmetry of nature in four spacetime dimensions began with the 1974 papers of Wess and Zumino.

Appendix A SU(6) Symmetry of Non-Relativistic Quark Models

This appendix will describe the way that an $SU(6)$ symmetry that relates particles of different spin arises in non-relativistic quark models. This has nothing directly to do with supersymmetry, but it provides the historical background for the Coleman–Mandula theorem, which is an essential input to the construction of general supersymmetry algebras in Sections 25.1 and 31.1.

In general, the Hamiltonian of the non-relativistic quark model could depend not only on positions and momenta but also on the spin and flavor operators $\sigma_i^{(n)}$ and $\lambda_A^{(n)}$, where the $\sigma_i^{(n)}$ (with $i = 1, 2, 3$) act on the spin indices of the nth quark as the Pauli matrices σ_i defined by Eq. (5.4.18), while the $\lambda_A^{(n)}$ (with $A = 1, 2, \ldots, 8$) act on the flavor indices of the nth quark as the Gell-Mann $SU(3)$ matrices λ_A defined by Eq. (19.7.2). (Where n refers to an antiquark, $\sigma_i^{(n)}$ and $\lambda_A^{(n)}$ act as the matrices $-\sigma_i^T$ and $-\lambda_A^T$ of the contragredient representations.) If we were to assume only that there is no spin-orbit coupling, so that the total orbital angular momentum L_i is separately conserved, then we could conclude only that the Hamiltonian commutes with the total spin and unitary spin

$$S_i \equiv \frac{1}{2}\sum_n \sigma_i^{(n)}, \qquad T_A \equiv \frac{1}{2}\sum_n \lambda_A^{(n)}, \tag{24.A.1}$$

as well as L_i. On the other hand, if we were to suppose that the Hamiltonian depends only on quark positions and momenta, and not at all on spins or quark flavors, then such a Hamiltonian would commute not only with the total orbital angular momentum L_i, but also with *each* of the operators $\sigma_i^{(n)}$ and $\lambda_A^{(n)}$. In between these two extremes there is the interesting possibility that in addition to commuting with the L_i, S_i, and T_A, the Hamiltonian also commutes with the operators

$$R_{iA} \equiv \frac{1}{2}\sum_n \pm\sigma_i^{(n)}\lambda_A^{(n)}, \tag{24.A.2}$$

where the sign is + or − for quarks and antiquarks, respectively.[4] The S_i, T_A, and R_{iA}

[4]The minus sign for antiquarks arises because the terms in R_{iA} for antiquarks must act on spin and flavor indices as the matrix $-(\sigma_i\lambda_A)^T = -(-\sigma_i^T)(-\lambda_A^T)$.

form the Lie algebra of the group $SU(6)$, with commutation relations

$$
\begin{aligned}
&[S_i, S_j] = i\sum_k \epsilon_{ijk} S_k, && [T_A, T_B] = i\sum_C f_{ABC} T_C,\\
&[S_i, T_A] = 0, && [S_i, R_{jA}] = i\sum_k \epsilon_{ijk} R_{kA},\\
&[T_A, R_{iB}] = i\sum_C f_{ABC} R_{iC}, &&\\
&[R_{Ai}, R_{Bj}] = i\delta_{ij}\sum_C f_{ABC} T_C + \frac{3}{2} i\delta_{AB}\sum_k \epsilon_{ijk} S_k &&\\
&\qquad + i\sum_{kC} \epsilon_{ijk} d_{ABC} R_{kC}. &&
\end{aligned}
\tag{24.A.3}
$$

Here f_{ABC} and d_{ABC} are respectively totally antisymmetric and totally symmetric numerical coefficients [14], with independent non-vanishing values given by

$$
\begin{aligned}
f_{123} = 1, \qquad f_{458} = f_{678} = \sqrt{3}/2,\\
f_{147} = f_{165} = f_{246} = f_{257} = f_{345} = f_{376} = 1/2,
\end{aligned}
\tag{24.A.4}
$$

and

$$
\begin{aligned}
d_{146} = d_{157} = -d_{247} = d_{256} = d_{344} = d_{355} = -d_{366} = -d_{377} = 1/2,\\
d_{118} = d_{228} = d_{338} = -d_{888} = 1/\sqrt{3},\\
d_{448} = d_{558} = d_{668} = d_{778} = -1/(2\sqrt{3}).
\end{aligned}
\tag{24.A.5}
$$

This is the symmetry that remains if we include a spin- and flavor-dependent two-body interaction in the Hamiltonian that commutes with R_{iA} as well as with the S_i and T_A. There are such interactions, given by linear combinations of two-body operators of the form

$$
H^{(nm)} \propto \left[1 \pm \sum_i \sigma_i^{(n)} \sigma_i^{(m)}\right] \left[\frac{2}{3} \pm \sum_A \lambda_A^{(n)} \lambda_A^{(m)}\right], \tag{24.A.6}
$$

where the sign $\pm$ is negative if one of the particles n, m is a quark and the other an antiquark and positive if they are both quarks or both antiquarks.

Of course, even in the non-relativistic quark model the $SU(6)$ symmetry is at best approximate. It is broken by spin-orbit and spin-spin forces, and also by the mass of the s quark, which reduces the flavor $SU(3)$ symmetry to the $SU(2)$ and $U(1)$ of isospin and hypercharge conservation. If we avoid the effects of this quark mass difference by restricting ourselves to hadrons built up from the light u and d quarks and antiquarks, then the only non-vanishing λ_A matrices are the λ_a with $a = 1, 2, 3$ (which for the u and d quarks are given by the Pauli matrices (5.4.18) (that in this context are conventionally called τ_a) and λ_8 (which is just the number $1/\sqrt{3}$ for the u and d quarks, and $-1/\sqrt{3}$ for the $\bar{u}$ and $\bar{d}$ antiquarks). The interaction (24.A.6) thus becomes

$$
H^{(nm)} \propto \left[1 \pm \sum_i \sigma_i^{(n)} \sigma_i^{(m)}\right] \left[1 \pm \sum_A \tau_A^{(n)} \tau_A^{(m)}\right]. \tag{24.A.7}
$$

Aside from quark number conservation, the remaining symmetry is then $SU(4)$, with generators S_i, T_a, and R_{ia} which commute with (24.A.7). This was proposed by Wigner [2] in 1937 as a symmetry of nuclear forces, though of course with protons and neutrons in place of u and d quarks. The interaction (24.A.7) is known in nuclear theory as a *Majorana potential* to distinguish it from an interaction that does not depend on spin or isospin, called a *Wigner potential*, or an interaction proportional to just the spin-dependent or just the isospin-dependent factor in (24.A.7), known respectively as a *Bartlett potential* and a *Heisenberg potential*.

It is amusing to note that, although in non-relativistic theories there is no theoretical barrier to symmetries like $SU(6)$ that act on spin as well as particle type, *there never was any experimental evidence for such an assumed $SU(6)$ symmetry of the non-relativistic quark model that is any better satisfied than the assumption of complete spin and flavor independence.* These assumptions are not the same; if the Hamiltonian of a system of N non-relativistic quarks and/or antiquarks is completely independent of spin and flavor, then its symmetry is $SU(6)^N$, not $SU(6)$. For instance, a two-particle interaction like (24.A.6) and various other multiparticle interactions break $SU(6)^N$ to $SU(6)$. Of course, all these symmetries are only approximate anyway. The question is whether $SU(6)$ is less badly broken than $SU(6)^N$?

This cannot be answered by studying the multiplet that contains the baryon octet, which consists of the nucleon and hyperons Λ, Σ, and Ξ. In the non-relativistic quark model these particles are interpreted as bound states of three quarks with zero orbital angular momentum. Because these states are color neutral, the wave function is completely antisymmetric in the suppressed color indices, and therefore it is completely symmetric under the combined interchange of spin and flavor. The baryon octet would therefore have to be put in the symmetric third-rank tensor representation **56** of $SU(6)$, which besides the baryon octet contains a spin 3/2 decuplet, which may be identified as the one consisting of the famous '3–3' resonance Δ and the $\Sigma(1385)$, $\Xi(1530)$, and Ω particles. (Numbers in parentheses give masses in MeV, where these are needed to distinguish the particles from others of the same isospin and strangeness but lower mass.) The $SU(6)$ symmetry leads to good predictions for the baryon magnetic moments: The quark charge operator is $q = e(\lambda_3/2 + \lambda_8/(2\sqrt{3}))$, so if quarks have the magnetic moments $3q/(2m_N)$ of Dirac particles of this charge and mass $m_N/3$, then the magnetic moment operator is

$$\mu_i \equiv 3\mu_N \left[\frac{1}{2}R_{i3} + \frac{1}{2\sqrt{3}}R_{i8}\right],$$

where $\mu_N \equiv e/(2m_N)$ is the nuclear magneton, and R_{iA} is defined by Eq. (24.A.2). It is straightforward to calculate the matrix elements of this symmetry generator between the members of the **56** multiplet, with the result that the magnetic moments for p, n, Λ, Σ^+, Σ^-, Ξ^-, and Ξ^0 in units of μ_N are respectively $+3$, -2, -1, $+3$, -1, -1, and -2, which may be compared with the corresponding experimental values $+2.79$, -1.91, -0.61, $+2.46$, -1.16, -0.65, and -1.25. The agreement is fair, and somewhat better (except for Σ^-) if we take the quark magnetic moment to be a little less than $3\mu_N$. Because of the symmetry of the three-quark wave function, nothing new is learned here if we assume that the Hamiltonian is completely independent of spin and flavor; the states of zero angular momentum would still have to fall in a multiplet consisting of $6 \cdot 7 \cdot 8/6! = 56$ members. In particular, the operator (24.A.6) has the same value 4 for any state of two quarks that is symmetric under simultaneous interchange of their spins and flavors, whether it is symmetric both under interchange of spins and interchange of flavors, or antisymmetric under both interchanges.

In order to decide whether $SU(6)$ is any better than $SU(6)^N$, it is more useful to study the mesons, which in the non-relativistic quark model are interpreted as bound states of a quark and antiquark. If the Hamiltonian of these states is completely independent of spin and flavor then its symmetry is $SU(6)^2$, and the meson states fall into its 36-dimensional $(\mathbf{6}, \overline{\mathbf{6}})$ representation, while for $SU(6)$ symmetry we could only say that the mesons belong to either of the two representations of $SU(6)$ contained in $\mathbf{6} \cdot \overline{\mathbf{6}}$: the adjoint representation **35** or the singlet representation. To be more specific, the **35** consists of an $SU(3)$ singlet with spin $S = 1$, an $SU(3)$ octet with $S = 0$, and an $SU(3)$ octet with $S = 1$, corresponding to the $SU(6)$ generators S_i, T_A, and R_{iA}, which is split from the $SU(3)$ singlet state with $S = 0$ by the interaction (24.A.6). Since all these assumptions are approximate anyway, the question in deciding whether $SU(6)$ symmetry is more accurate than complete spin and flavor independence is whether the splitting of the $SU(3)$ singlet $S = 0$ state from the other **35** states of the same orbital angular momentum is any greater than the splittings within the **35** supermultiplet.

For orbital angular momentum $L = 0$ the quark-antiquark states have negative parity P, and positive or negative charge-conjugation quantum number C (for self-charge-conjugate states) according to whether the total spin S is zero or one, respectively. (For an explanation, see Section 5.5.) The **35** therefore consists of a singlet with $J^{PC} = 1^{--}$, a 0^{-+} octet, and a 1^{--} octet, which may be identified respectively as: the $\phi(1020)$; the pseudoscalar octet π, η, K, and $\bar{K}$; and the vector octet ρ, ω, K^*, and $\bar{K}^*$. There is also a 0^{-+} $SU(3)$ singlet η' at 958 MeV, which can be regarded as the $SU(6)$ singlet. The splitting of this singlet from the particles in the **35** multiplet is not distinctly greater than the splittings within the **35** multiplet.

It may be argued that the $L = 0$ mesons do not provide a good test of the symmetries of the non-relativistic quark model, because they include the Goldstone bosons π, η, K, and $\bar{K}$, which become massless for zero u and d quark masses and are therefore not well described by this model. Therefore let us consider the quark-antiquark states with $L = 1$. These states have P positive and C positive or negative according to whether $S = 1$ or $S = 0$, so the p-wave **35** consists of: $SU(3)$ singlets with $S = 1$ and $J^{PC} = 0^{++}$, 1^{++}, 2^{++}, which may be identified as the $f_0(1370)$, the $f_1(1285)$, and the $f_2(1270)$; an $S = 0$ 1^{+-} octet identified as $h_1(1170)$, $b_1(1235)$, $K_1(1400)$, and $\bar{K}_1(1400)$; and $S = 1$ octets: a 0^{++} octet consisting of $f_0(980)$, $a_0(980)$, $K_0^+(1950)$, and $\bar{K}_0^+(1950)$; a 1^{++} octet consisting of $f_1(1420)$, $a_1(1260)$, $K_1^+(1650)$, and $\bar{K}_1^+(1650)$; and a 2^{++} octet consisting of $f_2(1430)$, $a_2(1320)$, $K_2^+(1980)$, and $\bar{K}_2^+(1980)$. In addition to these $\mathbf{35} \cdot 3$ states, there is a another particle with the right quantum numbers to be the p-wave $SU(6)$ singlet: the 1^{+-} isoscalar $h_1(1380)$. Of course, we could interchange the identifications of $h_1(1170)$ and $h_1(1380)$, or identify the $SU(3)$ singlet and octet isoscalar 1^{+-} states as orthogonal linear combinations of $h_1(1170)$ and $h_1(1380)$. The important point is that there are *two* of these 1^{+-} isoscalars, with no indication that one of them, belonging to an $SU(6)$ singlet, is more strongly split from the particles of the **35** than the particles of the **35** are split from each other. Here too, then, there is no evidence that $SU(6)$ symmetry is any more accurate than the stronger assumption of complete spin and flavor independence.

Appendix B The Coleman-Mandula Theorem

This appendix provides a proof of the celebrated theorem of Coleman and Mandula [5] that the only possible Lie algebra (as opposed to super-algebra) of symmetry generators consists of the generators P_μ and $J_{\mu\nu}$ of translations and homogeneous Lorentz transformations, together with possible internal symmetry generators, which commute with P_μ and $J_{\mu\nu}$ and act on physical states by multiplying them with spin-independent, momentum-independent Hermitian matrices.[5] By "symmetry generators" here is meant any Hermitian operators: that commute with the S-matrix; whose commutators are also symmetry generators; which take one-particle states into one-particle states; and whose action on multiparticle states is the direct sum of their action on one-particle states (as in Eq. (24.B.1)). A further technical requirement will be added when needed later. Apart from the general principles of relativistic quantum mechanics described in Chapters 2 and 3, the only other assumptions needed in this proof are:

Assumption 1 *For any M there are only a finite number of particle types with mass less than M.*

Assumption 2 *Any two-particle state undergoes some reaction at almost all energies (that is, at all energies except perhaps an isolated set).*

Assumption 3 *The amplitudes for elastic two-body scattering are analytic functions of the scattering angle at almost all energies and angles.*[6]

It is not necessary to assume that the S-matrix is governed by a local quantum field theory. The proof presented here is somewhat rearranged and streamlined, and it spells out some steps that Coleman and Mandula left to the reader.

It is convenient to start by proving this theorem for the subalgebra consisting of those symmetry generators B_α that commute with the four-momentum operator P_μ. (This part of the theorem is of some interest in itself; it rules out symmetries in relativistic theories

[5]As we shall see, in theories with only massless particles there is also the possibility that in addition to the generators P_μ and $J_{\mu\nu}$ there are additional generators D and K_μ that fill out the Lie algebra of the conformal group [15].

[6]Strictly speaking, this assumption is not satisfied in theories with infrared divergences such as quantum electrodynamics, where, as shown in Section 13.3, the S-matrix element for any one scattering process involving charged particles actually vanishes, except for elastic forward scattering. In Abelian gauge theories like electrodynamics this problem can be avoided by applying the Coleman–Mandula theorem to the theory with a fictitious gauge boson mass, and then working only with 'infrared-safe' quantities like masses and suitably integrated cross-sections that are finite in the limit of zero gauge boson mass. There is no problem in non-Abelian gauge theories like quantum chromodynamics, in which all massless particles are trapped— symmetries if unbroken would only govern S-matrix elements for gauge-neutral bound states, like the mesons and baryons in quantum chromodynamics. As far as I know, the Coleman–Mandula theorem has not been proved for non-Abelian gauge theories with untrapped massless particles, like quantum chromodynamics with many quark flavors.

that act like the $SU(6)$ symmetry of the non-relativistic quark model.) The action of such symmetry generators on multiparticle states is given by

$$\begin{aligned} B_\alpha \| pm, qn, \ldots \rangle = & \sum_{m'} (b_\alpha(p))_{m'm} \| pm', qn, \ldots \rangle \\ & + \sum_{n'} (b_\alpha(q))_{n'n} \| pm, qn', \ldots \rangle + \cdots , \end{aligned} \tag{24.B.1}$$

where m, n, etc. are discrete indices labelling spin z-components and particle type for particles of a definite mass $\sqrt{-p_\mu p^\mu}$, and the $b_\alpha(p)$ are finite Hermitian matrices, which define the action of the B_α on one-particle states.

Now, we can see from Eq. (24.B.1) that the mapping that takes the B_α into $b_\alpha(p)$ for some fixed p is a homomorphism in the sense that the commutation relations

$$[B_\alpha, B_\beta] = i \sum_\gamma C^\gamma{}_{\alpha\beta} B_\gamma \tag{24.B.2}$$

are also satisfied by the Hermitian matrices $b_\alpha(p)$:

$$[b_\alpha(p), b_\beta(p)] = i \sum_\gamma C^\gamma{}_{\alpha\beta} b_\gamma(p). \tag{24.B.3}$$

A well-known theorem proved in Section 15.2 tells us that any Lie algebra of finite Hermitian matrices like $b_\alpha(p)$ must be a direct sum of a compact semi-simple Lie algebra and $U(1)$ algebras. However, we cannot immediately apply this result to the operator algebras B_α because the homomorphism between the operators B_α and matrices $b_\alpha(p)$ is not necessarily an isomorphism. For it to be an isomorphism would require also that whenever $\sum_\alpha c^\alpha b_\alpha(p) = 0$ for some coefficients c^α and momentum p, then $\sum_\alpha c^\alpha b_\alpha(k) = 0$ for all momenta k, which is equivalent to the condition $\sum_\alpha c^\alpha B_\alpha = 0$.

Instead of considering the homomorphism that maps the B_α into the one-particle matrices $b_\alpha(p)$, Coleman and Mandula considered the homomorphism that maps the B_α into the matrices that define the action of B_α on two-particle states with fixed four-momenta p and q:

$$(b_\alpha(p, q))_{m'n', mn} = (b_\alpha(p))_{m'm} \delta_{n'n} + (b_\alpha(q))_{n'n} \delta_{m'm}. \tag{24.B.4}$$

The invariance of the S-matrix for the elastic or quasi-elastic scattering of two particles with four-momenta p and q into two particles with momenta p' and q', with masses $\sqrt{-p'_\mu p'^\mu} = \sqrt{-p_\mu p^\mu}$ and $\sqrt{-q'_\mu q'^\mu} = \sqrt{-q_\mu q^\mu}$, yields the condition

$$b_\alpha(p', q') S(p', q'; p, q) = S(p', q'; p, q) b_\alpha(p, q). \tag{24.B.5}$$

Here $S(p', q'; p, q)$ is a matrix of the same dimensionality as $b(p, q)$ and $b(p', q')$, defined in terms of the connected S-matrix elements $S(pm, qn \to p'm', q'n')$ by

$$S(pm, qn \to p'm', q'n') \equiv \delta^4(p' + q' - p - q) \big(S(p', q'; p, q)\big)_{m'n', mn}. \tag{24.B.6}$$

According to Assumption 2 and the optical theorem (see Section 3.6), for almost any choice of p and q the elastic scattering amplitude is non-vanishing in the forward direction, and

Assumption 3 then tells us that the matrix $S(p',q';p,q)$ is non-singular for almost all p' and q' on the same mass shells and satisfying the conservation condition $p'+q'=p+q$, so for almost all such four-momenta Eq. (24.B.5) is a *similarity transformation*.

It follows then that if $\sum_\alpha c^\alpha b_\alpha(p,q)=0$ for almost any fixed four-momenta p and q, then $\sum_\alpha c^\alpha b_\alpha(p',q')=0$ for almost any four-momenta p' and q' on the same mass shells that satisfy $p'+q'=p+q$. Unfortunately, this does not tell us that $\sum_\alpha c^\alpha b_\alpha(p')$ and $\sum_\alpha c^\alpha b_\alpha(q')$ vanish, but only that these matrices are proportional to the unit matrix (with opposite coefficients). To do better, it is necessary to consider not the $b_\alpha(p)$ or $b_\alpha(p,q)$, but their traceless parts.

One immediate consequence of Eq. (24.B.5) is that

$$\mathrm{Tr}\, b_\alpha(p',q') = \mathrm{Tr}\, b_\alpha(p,q). \tag{24.B.7}$$

With Eq. (24.B.4), this tells us that

$$\begin{aligned} &N\big(\sqrt{-q_\mu q^\mu}\big)\,\mathrm{tr}\, b_\alpha(p') + N\big(\sqrt{-p_\mu p^\mu}\big)\,\mathrm{tr}\, b_\alpha(q') \\ &\quad = N\big(\sqrt{-q_\mu q^\mu}\big)\,\mathrm{tr}\, b_\alpha(p) + N\big(\sqrt{-p_\mu p^\mu}\big)\,\mathrm{tr}\, b_\alpha(q), \end{aligned} \tag{24.B.8}$$

where $N(m)$ is the multiplicity[7] of particle types with mass m, and the lower case t in 'tr' indicates a sum over one-particle rather than two-particle labels. In order for this to be satisfied for almost all mass-shell four-momenta for which $p'+q'=p+q$, it is necessary that the function $\mathrm{tr}\, b_\alpha(p)/N\big(\sqrt{-p_\mu p^\mu}\big)$ be linear[8] in p:

$$\frac{\mathrm{tr}\, b_\alpha(p)}{N\big(\sqrt{-p_\mu p^\mu}\big)} = a_\alpha^\mu p_\mu, \tag{24.B.9}$$

with a_α^μ independent of p (and of everything else but the displayed indices.) We may define new symmetry generators by subtracting terms linear in the momentum operator P_μ:

$$B_\alpha^\# \equiv B_\alpha - a_\alpha^\mu P_\mu, \tag{24.B.10}$$

which according to Eq. (24.B.9) are represented on one-particle states by the traceless matrices

$$\big(b_\alpha^\#(p)\big)_{n'n} = \big(b_\alpha(p)\big)_{n'n} - \frac{\mathrm{tr}\, b_\alpha(p)}{N\big(\sqrt{-p_\mu p^\mu}\big)}\delta_{n'n}. \tag{24.B.11}$$

Because P_μ commutes with B_α and the unit matrix commutes with everything, the commutators of the $B_\alpha^\#$ are the same as those of the B_α, and the commutators of the $b_\alpha^\#(p)$ are the same as those of the $b_\alpha(p)$:

$$[B_\alpha^\#, B_\beta^\#] = i\sum_\gamma C^\gamma{}_{\alpha\beta} B_\gamma = i\sum_\gamma C^\gamma{}_{\alpha\beta}\big[B_\gamma^\# + a_\gamma^\mu P_\mu\big], \tag{24.B.12}$$

[7] These multiplicity factors were not shown explicitly by Coleman and Mandula. They are needed in justifying a step that Coleman and Mandula made without explanation, that of defining the symmetry generators $B_\alpha^\#$ with traceless kernels.

[8] A constant term is easily seen to be ruled out in Eq. (24.B.9) by the existence of processes in which the number of particles is not conserved, processes which are inevitable in any relativistic quantum theory satisfying the cluster decomposition principle. Even if we considered only two-particle processes and did not use this argument, a constant term in Eq. (24.B.9) would only amount to a change in the action of internal symmetries on physical states.

$$[b^{\#}_\alpha(p), b^{\#}_\beta(p)] = i\sum_\gamma C^\gamma{}_{\alpha\beta} b_\gamma(p) = i\sum_\gamma C^\gamma{}_{\alpha\beta}\big[b^{\#}_\gamma(p) + a^\mu_\gamma p_\mu\big]. \tag{24.B.13}$$

Also, Eq. (24.B.13) and the fact that commutators of the finite matrices $b^{\#}_\alpha(p)$ have zero trace[9] imply that $\sum_\gamma C^\gamma{}_{\alpha\beta} a^\mu_\gamma = 0$, and using this in Eq. (24.B.12) then shows that the $B^{\#}_\alpha$ satisfy the same commutation relations as the B_α:

$$[B^{\#}_\alpha, B^{\#}_\beta] = i\sum_\gamma C^\gamma{}_{\alpha\beta} B^{\#}_\gamma. \tag{24.B.14}$$

Because $B^{\#}_\alpha$ is a symmetry generator, the scattering amplitude satisfies

$$b^{\#}_\alpha(p', q') S(p', q'; p, q) = S(p', q'; p, q) b^{\#}_\alpha(p, q), \tag{24.B.15}$$

where the $b^{\#}_\alpha(p,q)$ are the matrices representing the $B^{\#}_\alpha$ on the two-particle states

$$\big(b^{\#}_\alpha(p,q)\big)_{m'n',mn} = \big(b^{\#}_\alpha(p)\big)_{m'm}\delta_{n'n} + \big(b^{\#}_\alpha(q)\big)_{n'n}\delta_{m'm} \tag{24.B.16}$$

and satisfy the same commutation relations as the $B^{\#}_\alpha$:

$$[b^{\#}_\alpha(p,q), b^{\#}_\beta(p,q)] = i\sum_\gamma C^\gamma{}_{\alpha\beta} b^{\#}_\gamma(p,q). \tag{24.B.17}$$

The advantage of dealing with these two-particle matrices is that, since $S(p', q'; p, q)$ is a non-singular matrix, it follows that if $\sum_\alpha c^\alpha b^{\#}_\alpha(p,q) = 0$ for some fixed mass-shell four-momenta p and q, then $\sum_\alpha c^\alpha b^{\#}_\alpha(p', q') = 0$ for almost all p' and q' on the same respective mass shells with $p' + q' = p + q$. Because we are dealing now with traceless matrices, this tells us that

$$\sum_\alpha c^\alpha b^{\#}_\alpha(p') = \sum_\alpha c^\alpha b^{\#}_\alpha(q') = 0. \tag{24.B.18}$$

We would like to conclude from this that $\sum_\alpha c^\alpha b^{\#}_\alpha(k) = 0$ for *all* mass-shell four-momenta k, but so far we have proved only that $\sum_\alpha c^\alpha b^{\#}_\alpha(p') = 0$ for almost all of those p' for which $q' = p + q - p'$ as well as p' is on the mass shell (and correspondingly for q'.) To get around this limitation, we can use a trick of Coleman and Mandula, noting that if $\sum_\alpha c^\alpha b^{\#}_\alpha(p,q) = 0$ then (24.B.18) and (24.B.16) together yield

$$\sum_\alpha c^\alpha b^{\#}_\alpha(p, q') = 0,$$

so that, according to Eq. (24.B.15),

$$\sum_\alpha c^\alpha b^{\#}_\alpha(k, p + q' - k) = 0,$$

[9] This is one of the places where we use Assumption 1, without which commutators need not have zero trace. Also, at this point it is crucial that we are dealing with commutation rather than anticommutation relations, since the unit matrix does not anticommute with other matrices, and anticommutators of finite matrices do not necessarily have zero trace.

and therefore

$$\sum_\alpha c^\alpha b^\#_\alpha(k) = 0, \tag{24.B.19}$$

for almost all mass-shell four-momenta k for which $p + q' - k$ is also on the mass shell. Now, the conditions that q' and $p + q - q'$ be on the mass shell leave two parameters free in q', so that we have enough freedom in choosing q' that the condition that $p + q' - k$ is on the mass shell leaves us free to choose k to be anything we like, at least within a finite volume of momentum space. This volume can be adjusted to be as large as we like by taking p and q sufficiently large, so if $\sum_\alpha c^\alpha b^\#_\alpha(p, q) = 0$ for some fixed mass-shell four-momenta p and q then $\sum_\alpha c^\alpha b^\#_\alpha(k) = 0$ for almost all mass-shell four-momenta k. But then if $\sum_\alpha c^\alpha b^\#_\alpha(k_0) \neq 0$ for some particular mass-shell four-momentum k_0, a scattering process in which particles with four-momenta k_0 and k scatter into particles with four-momenta k' and k'' will be forbidden by the symmetry generated by $\sum_\alpha c^\alpha B^\#_\alpha$ for almost all k, k', and k'', in contradiction with our assumption about the analyticity of the scattering amplitude. We conclude then that if $\sum_\alpha c^\alpha b^\#_\alpha(p, q) = 0$ for some fixed mass-shell four-momenta p and q then $\sum_\alpha c^\alpha b^\#_\alpha(k) = 0$ for all k, and therefore $\sum_\alpha c^\alpha B^\#_\alpha = 0$, so that the mapping that takes B_α into $b^\#_\alpha(p, q)$ is an isomorphism.

One immediate consequence is that, since the number of independent matrices $b^\#_\alpha(p, q)$ cannot exceed $N\big(\sqrt{-p_\mu p^\mu}\big)N\big(\sqrt{-q_\mu q^\mu}\big)$, there can be at most a finite number of independent symmetry generators B_α. As emphasized by Coleman and Mandula, in proving their theorem it is not necessary to make an independent assumption that the symmetry algebra is finite-dimensional.

The theorems of Section 15.2 tell us further that a Lie algebra of finite Hermitian matrices like $b^\#_\alpha(p, q)$ for fixed p and q is at most the direct sum of a semi-simple compact Lie algebra and some number of $U(1)$ Lie algebras. We have seen that this Lie algebra is isomorphic to that of the symmetry generators $B^\#_\alpha$, so the $B^\#_\alpha$ must also span the direct sum of at most a compact semi-simple Lie algebra and $U(1)$ Lie algebras.

Let us first dispose of the $U(1)$ Lie algebras. For any pair of mass-shell momenta p and q we can find a Lorentz generator J that leaves both p and q invariant. (If p and q are lightlike and parallel then take J to generate rotations around the common direction of $\mathbf{p}$ and $\mathbf{q}$. Otherwise, $p + q$ will be timelike, so we can take J to generate rotations around the common direction of $\mathbf{p}$ and $\mathbf{q}$ in the center-of-mass frame in which $\mathbf{p} = -\mathbf{q}$.) We can choose the basis of two-particle states to diagonalize J, so that

$$J\|pm, qn\rangle = \sigma(m, n)\|pm, qn\rangle. \tag{24.B.20}$$

Now, P_μ commutes with all $B^\#_\alpha$, and $[J, P_\mu]$ is a linear combination of components of P_μ, so P_μ commutes with all $[J, B^\#_\alpha]$, and therefore the symmetry generator $[J, B^\#_\alpha]$ must be a linear combination of the B_β, which by definition form a complete set of symmetry generators that commute with P_μ. More specifically, since the matrices representing a commutator of symmetry generators are necessarily traceless, $[J, B^\#_\alpha]$ has to be a linear combination of the $B^\#_\beta$. But any $U(1)$ generator $B^\#_i$ (taken Hermitian) in the algebra of the $B^\#_\beta$ would have to commute with all the $B^{\#}_{\beta}$, and hence in particular must commute with $[J, B^{\#}_i]$:

$$[B^{\#}_i, [J, B^{\#}_i]] = 0.$$

Taking the expectation value of this double commutator in the two-particle basis in which J is diagonal, we have

$$0 = \sum_{m',n'} \left(\sigma(m', n') - \sigma(m, n)\right) \left|\left(b^{\#}_i(p, q)\right)_{m'n',mn}\right|^2 , \tag{24.B.21}$$

for any m and n. The indices run over a finite range, so if there were any σ for which there existed an m and n with $\sigma(m, n) = \sigma$ and an m' and n' with $\sigma(m', n') \neq \sigma$, with $\left(b^{\#}_i(p, q)\right)_{m'n',mn} \neq 0$, then there would have to be a *smallest* such σ, in which case the right-hand side of Eq. (24.B.21) for this m and n would be positive-definite, contradicting Eq. (24.B.21). We conclude that $\left(b^{\#}_i(p, q)\right)_{m'n',mn}$ must vanish for all m, n, m', and n' for which $\sigma(m', n') \neq \sigma(m, n)$. Because the algebra of the $b^{\#}_i(p, q)$ is isomorphic to that of the $B^{\#}_i$, this means that each of the $U(1)$ generators $B^{\#}_i$ commutes with J. Since we can choose $p+q$ to be in any timelike direction, it follows that each of the $U(1)$ generators $B^{\#}_i$ commutes with all the generators $J_{\mu\nu}$ of the homogeneous Lorentz group. The fact that they commute with what we called 'boosts' in Section 2.5 implies that the $\left(b^{\#}_i(p)\right)_{n'n}$ are independent of three-momentum, and the fact that they commute with rotations implies that the $\left(b^{\#}_i(p)\right)_{n'n}$ act as unit matrices on spin indices, so these generators are the generators of an ordinary internal symmetry.

This leaves the $B^{\#}_{\alpha}$ that generate a semi-simple compact Lie algebra. The argument of Section 24.1 (a somewhat more explicit version of the reasoning given by Coleman and Mandula) tells us that the generators of the semi-simple compact part of the Lie algebra commute with Lorentz transformations and, as shown for the $U(1)$ generators, this means that they too are the generators of internal symmetries. We have thus shown that the symmetry generators B_{α} that commute with P_{μ} are either internal symmetry generators or linear combinations of the components of P_{μ} itself.

Next, we must take up the possibility of symmetry generators that do not commute with the momentum operator. The action of a general symmetry generator A_{α} on a one-particle state $\|p, n\rangle$ of four-momentum p would be

$$A_{\alpha}\|p, n\rangle = \sum_{n'} \int d^4p' \left(\mathcal{A}_{\alpha}(p', p)\right)_{n'n} \|p', n'\rangle, \tag{24.B.22}$$

where n and n' are again discrete indices labelling both spin z-components and particle types. Of course, the kernel $\mathcal{A}_{\alpha}(p', p)$ must vanish unless both p and p' are on the mass shell. We shall show first that the $\mathcal{A}_{\alpha}(p', p)$ vanishes for any $p' \neq p$.

For this purpose, note that if A_{α} is a symmetry generator, then so is

$$A^f_{\alpha} \equiv \int d^4x \, \exp(iP \cdot x) A_{\alpha} \exp(-iP \cdot x) f(x), \tag{24.B.23}$$

where P_μ is the four-momentum operator, and $f(x)$ is a function that can be chosen as we like. Acting on a one-particle state, this gives

$$A^f_\alpha \| p, n\rangle = \sum_{n'} \int d^4p'\, \widetilde{f}(p'-p) \big(\mathcal{A}_\alpha(p',p)\big)_{n'n} \| p', n'\rangle, \tag{24.B.24}$$

where $\widetilde{f}$ is the Fourier transform

$$\widetilde{f}(k) \equiv \int d^4x\, \exp(ix\cdot k) f(x). \tag{24.B.25}$$

Suppose that there is some pair of mass-shell four-momenta p and $p+\Delta$ with $\Delta \neq 0$ such that $\mathcal{A}(p+\Delta, p) \neq 0$. Generic mass-shell four-momenta q, p', and q' with $p'+q' = p+q$ will not have $q+\Delta$ or $p'+\Delta$ or $q'+\Delta$ on the mass shell. If we take $\widetilde{f}(k)$ to vanish outside a sufficiently small region around Δ, then A^f_α will annihilate all one-particle states with four-momenta q, p', and q', but not the one-particle states with four-momentum p, so such a symmetry would forbid any scattering process in which any particles of momenta p and q go into any particles of momenta p' and q', in contradiction with the consequence of Assumptions 2 and 3 that there is some scattering at almost all energies and angles.

This result does not mean that any symmetry generator A_α must commute with P_μ, because the kernels $\mathcal{A}_\alpha(p',p)$ may include terms proportional to derivatives of $\delta^4(p'-p)$ as well as terms proportional to $\delta^4(p'-p)$ itself. To deal with this possibility, Coleman and Mandula made the 'ugly technical assumption' that the kernels $\mathcal{A}_\alpha(p',p)$ are *distributions*, which means that each can contain at most a finite number D_α of derivatives of $\delta^4(p'-p)$. To put this another way, each symmetry generator A_α is assumed to act on one-particle states as a polynomial of order D_α in the derivatives $\partial/\partial p_\mu$, with matrix coefficients that at this point are allowed to depend on momentum and spin. To use the above results for symmetry generators that commute with the momentum operators, Coleman and Mandula considered the D_α-fold commutator of momentum operators with A_α

$$B_\alpha^{\mu_1\cdots\mu_{D_\alpha}} \equiv [P^{\mu_1}, [P^{\mu_2}, \ldots, [P^{\mu_{D_\alpha}}, A_\alpha]]\ldots]]. \tag{24.B.26}$$

The matrix elements of the commutators of $B_\alpha^{\mu_1\cdots\mu_{D_\alpha}}$ with P^μ between states with four-momenta p' and p are proportional to $D_\alpha+1$ factors of $p'-p$, times a polynomial of order D_α in momentum derivatives acting on $\delta^4(p'-p)$, and therefore vanish. Since the generators $B_\alpha^{\mu_1\cdots\mu_{D_\alpha}}$ commute with the momentum operators, according to the results obtained so far they act on one-particle states with matrices of the form

$$b_\alpha^{\mu_1\cdots\mu_{D_\alpha}}(p) = b_\alpha^{\#\mu_1\cdots\mu_{D_\alpha}} + a_\alpha^{\mu\mu_1\cdots\mu_{D_\alpha}} p_\mu \mathbf{1}, \tag{24.B.27}$$

where the $b_\alpha^{\#\mu_1\cdots\mu_{D_\alpha}}$ are momentum-independent traceless Hermitian matrices generating an ordinary internal symmetry algebra, and the $a_\alpha^{\mu\mu_1\cdots\mu_{D_\alpha}}$ are momentum-independent numerical constants, with both $b_\alpha^{\#\mu_1\cdots\mu_{D_\alpha}}$ and $a_\alpha^{\mu\mu_1\cdots\mu_{D_\alpha}}$ symmetric in the indices $\mu_1\cdots\mu_{D_\alpha}$. Also, even though the A_α do not necessarily commute with P_μ, they cannot take one-particle states off the mass shell, so since Assumption 1 requires that the mass-squared operator

$-P_\mu P^\mu$ has only discrete eigenvalues, the A_α must commute with $-P_\mu P^\mu$. It follows in particular that for $D \geq 1$

$$0 = [P^\mu P_\mu, [P^{\mu_2}, \ldots, [P^{\mu_{D_\alpha}}, A_\alpha]] \ldots]] = 2P_{\mu_1} B_\alpha^{\mu_1 \cdots \mu_{D_\alpha}},$$

so that

$$0 = p_{\mu_1} b_\alpha^{\mu_1 \cdots \mu_{D_\alpha}}(p). \tag{24.B.28}$$

As long as the theory contains massive particles this must be satisfied for p in any timelike direction, so for $D_\alpha \geq 1$

$$b_\alpha^{\#\mu_1 \cdots \mu_{D_\alpha}} = 0, \tag{24.B.29}$$

and

$$a_\alpha^{\mu\mu_1 \cdots \mu_{D_\alpha}} = -a_\alpha^{\mu_1 \mu \cdots \mu_{D_\alpha}}. \tag{24.B.30}$$

But for $D_\alpha \geq 2$ Eq. (24.B.30) together with the symmetry of $a_\alpha^{\mu\mu_1 \cdots \mu_{D_\alpha}}$ in the indices $\mu_1 \cdots \mu_{D_\alpha}$ would require that $a_\alpha^{\mu\mu_1 \cdots \mu_{D_\alpha}} = 0$. (For then

$$\begin{aligned} a_\alpha^{\mu\mu_1\mu_2 \cdots \mu_{D_\alpha}} &= a_\alpha^{\mu\mu_2\mu_1 \cdots \mu_{D_\alpha}} = -a_\alpha^{\mu_2\mu\mu_1 \cdots \mu_{D_\alpha}} = -a_\alpha^{\mu_2\mu_1\mu \cdots \mu_{D_\alpha}} \\ &= a_\alpha^{\mu_1\mu_2\mu \cdots \mu_{D_\alpha}} = a_\alpha^{\mu_1\mu\mu_2 \cdots \mu_{D_\alpha}} = -a_\alpha^{\mu\mu_1\mu_2 \cdots \mu_{D_\alpha}}.) \end{aligned}$$

We are left with at most two kinds of non-vanishing symmetry generator: those with $D_\alpha = 0$, for which the generator A_α commutes with P_μ and therefore must be either an internal symmetry generator or some linear combination of the P_μ; and those with $D_\alpha = 1$, in which case

$$[P^\nu, A_\alpha] = a_\alpha^{\mu\nu} P_\mu, \tag{24.B.31}$$

with $a_\alpha^{\mu\nu}$ some numerical constants antisymmetric in μ and ν. Eq. (24.B.31) requires that

$$A_\alpha = -\frac{1}{2} i a_\alpha^{\mu\nu} J_{\mu\nu} + B_\alpha, \tag{24.B.32}$$

where $J_{\mu\nu}$ is the generator of proper Lorentz transformations, which according to Eq. (2.4.13) satisfies $[P^\nu, J^{\rho\sigma}] = -i\eta^{\nu\rho}P^\sigma + i\eta^{\nu\sigma}P^\rho$, and B_α commutes with P_μ. Since A_α and $J_{\mu\nu}$ are symmetry generators, so is B_α, which must therefore be a linear combination of internal symmetry generators and/or the components of P_μ. Eq. (24.B.32) therefore completes the proof of the Coleman–Mandula theorem.

* * *

In theories with only massless particles Eq. (24.B.30) does not necessarily follow from Eq. (24.B.28); since $p_\mu p^\mu = 0$, we can also have

$$a_\alpha^{\mu\mu_1 \cdots \mu_{D_\alpha}} + a_\alpha^{\mu_1\mu \cdots \mu_{D_\alpha}} \propto \eta^{\mu\mu_1}. \tag{24.B.33}$$

In this case the symmetry algebra consists of internal symmetries plus the algebra of the conformal group, which is spanned by generators K^μ and D together with the generators $J^{\mu\nu}$ and P^μ of the Poincaré group. The commutation relations are

$$\begin{aligned} &[P^\mu, D] = iP^\mu, && [K^\mu, D] = -iK^\mu, \\ &[P^\mu, K^\nu] = 2i\eta^{\mu\nu}D + 2iJ^{\mu\nu}, && [K^\mu, K^\nu] = 0, \\ &[J^{\rho\sigma}, K^\mu] = i\eta^{\mu\rho}K^\sigma - i\eta^{\mu\sigma}K^\rho, && [J^{\rho\sigma}, D] = 0, \end{aligned} \tag{24.B.34}$$

together with the commutation relations (2.4.12)–(2.4.14) of the Poincaré algebra

$$\begin{aligned} i[J^{\mu\nu}, J^{\rho\sigma}] &= \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\sigma\mu}J^{\rho\nu} + \eta^{\sigma\nu}J^{\rho\mu}, \\ i[P^{\mu}, J^{\rho\sigma}] &= \eta^{\mu\rho}P^{\sigma} - \eta^{\mu\sigma}P^{\rho}, \\ [P^{\mu}, P^{\rho}] &= 0. \end{aligned} \tag{24.B.35}$$

The infinitesimal group element

$$U(1+\omega, \epsilon, \lambda, \rho) = 1 + (i/2)J_{\mu\nu}\omega^{\mu\nu} + iP_{\mu}\epsilon^{\mu} + i\lambda D + iK_{\mu}\rho^{\mu} \tag{24.B.36}$$

induces the infinitesimal spacetime transformation

$$x^{\mu} \longrightarrow x^{\mu} + \omega^{\mu\nu}x_{\nu} + \epsilon^{\mu} + \lambda x^{\mu} + \rho^{\mu}x^{\nu}x_{\nu} - 2x^{\mu}\rho^{\nu}x_{\nu}. \tag{24.B.37}$$

These are the most general infinitesimal spacetime transformations that leave the light-cone invariant.

Problems

1. Show that the most general symmetry algebra allowed under the assumptions of the Coleman–Mandula theorem in the case where all particles are massless consists of internal symmetry generators plus either the Poincaré algebra or the conformal algebra (24.B.34), (24.B.35).

2. Show that the Gervais–Sakita action (24.2.5) is invariant under the worldsheet supersymmetry transformation (24.2.7).

3. Calculate the change in the Wess–Zumino Lagrangian density (24.2.9) under the spacetime supersymmetry transformation (24.2.8).

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